Menu Recommendation System: Applying Apriori Association Rules and Collaborative Filtering for Food Suggestion and Automatic Combo Generation

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Abstract

Digital restaurant platforms generate a large volume of behavioral data from search logs, menu views, carts, orders, ratings, and repeated purchases. However, many menu systems still display static categories or manually configured promotions, making it difficult to personalize recommendations, create effective food combos, or manage customer anxiety after ordering. This paper presents CrownDine, a menu recommendation and operational management system that applies association rule mining with the Apriori algorithm, collaborative filtering, and kitchen display system (KDS) status updates to support the full customer journey. Apriori is used to discover frequent co-purchase patterns and generate interpretable rules such as dish-to-side or dish-to-drink associations. Collaborative filtering personalizes suggestions by learning similarities among users or menu items from historical interactions. Beyond algorithmic recommendation, the proposed system is framed as a dual-phase cognitive support tool: digital nudging reduces decision overload during menu selection, while real-time order visibility reduces waiting anxiety after checkout. The system design includes transaction preprocessing, frequent itemset mining, rating and interaction matrix construction, hybrid recommendation scoring, combo validation, KDS-based order status updates, and evaluation using support, confidence, lift, precision, recall, ranking quality, perceived usefulness, and customer waiting satisfaction metrics.

Keywords: Menu Recommendation, Apriori Algorithm, Association Rule Mining, Collaborative Filtering, Food Combo Generation, Restaurant Analytics

1 Introduction

Restaurant customers increasingly interact with digital menus through mobile applications, QR-code ordering systems, and online food delivery platforms. These interactions produce valuable behavioral signals, including dish views, search keywords, cart additions, order histories, ratings, time-of-day preferences, and repeated purchases. When properly analyzed, such data can help restaurants recommend relevant dishes, suggest complementary items, and create attractive combos that match actual customer behavior rather than relying only on manual menu engineering.

Traditional menu systems often organize dishes by static categories such as appetizers, main courses, drinks, and desserts. Although this structure is easy to browse, it does not adapt to individual preferences or reveal hidden purchasing patterns. More importantly, large menus can create decision overload: customers must compare many dishes, prices, categories, and meal contexts before making a choice. Excessive choice can increase cognitive effort, produce decision fatigue, and encourage default choices or choice deferral. For example, customers who order pasta may frequently add red wine, while customers who choose fried chicken may often purchase soft drinks or specific side dishes. If these patterns are discovered automatically, the system can recommend add-ons and generate combo packages that improve convenience for customers and increase average order value for restaurants.

This paper focuses on the topic **Menu Recommendation System: Applying Apriori Association Rules and Collaborative Filtering for Food Suggestion and Automatic Combo Generation Based on User Behavior**. The extended CrownDine perspective treats the system not only as a recommendation engine but also as a behavioral support system that addresses two customer-experience frictions: decision overload during dish selection and waiting anxiety after ordering. The proposed system combines four complementary components:

1. **Apriori Association Rule Mining:** Discovering frequent itemsets and co-purchase rules from transaction data to identify dishes that are commonly ordered together.
2. **Collaborative Filtering:** Using user–item interaction patterns to recommend dishes based on similarities between users or similarities between menu items.
3. **Hybrid Recommendation Ranking:** Combining association-rule confidence, lift, user preference scores, item popularity, and contextual signals to rank dish suggestions.
4. **Automatic Combo Generation:** Creating dish bundles from frequent itemsets and validating them using business constraints such as category balance, price range, inventory availability, and profit margin.
5. **KDS-Based Waiting Support:** Displaying real-time order preparation status to reduce uncertainty, increase perceived control, and improve the waiting experience.

Based on this research gap, the study is guided by the following research questions:

RQ1: How do recommendation algorithms, including Apriori and collaborative filtering, operate as digital nudges that reduce customer decision overload during the menu-selection stage?

RQ2: How does real-time order-status synchronization through the kitchen display system affect customers' cognitive control and waiting anxiety during the post-order stage?

RQ3: How can automatically generated food combos based on user behavior balance restaurant business constraints with personalized customer experience?

RQ4: How does CrownDine's dual-phase cognitive support framework reduce perceived friction across the customer journey, from dish selection to order waiting?

The remainder of this paper is organized as follows. Section 2 reviews related studies on menu recommendation, digital nudging, waiting psychology, association rule mining, collaborative filtering, and food combo design. Section 3 presents the proposed methodology and system architecture in relation to the four research questions. Section 4 discusses expected benefits, limitations, and the behavioral implications of the dual-phase cognitive support framework, and Section 5 concludes the paper.

2 Related Work

The design of a menu recommendation system is closely related to research in consumer behavior, recommender systems, and restaurant analytics. This section summarizes the main foundations relevant to association-rule-based suggestions and collaborative filtering.

2.1 Digital Menu Recommendation and Consumer Behavior

Digital food ordering platforms influence customer decisions by changing how menu information is presented, filtered, and promoted. Prior studies on restaurant experiences and electronic word-of-mouth show that information quality, credibility, and usefulness affect customer choice intentions [1]. In online ordering environments, perceived convenience, price transparency, and delivery information also shape customer behavior [2]. These findings suggest that recommendation systems should not only display popular dishes but also provide relevant, timely, and trustworthy suggestions that reduce search effort.

Customer satisfaction in restaurants is influenced by food quality, perceived value, and service experience [3, 4]. A recommendation system can support these factors by helping customers find dishes that match their preferences, budget, and meal context. In addition, digital nudging research shows that interface-level interventions can influence food choices without reducing satisfaction [5]. Therefore, recommendations and combo offers should be designed as helpful decision support rather than intrusive advertisements.

2.2 Digital Nudging and Cognitive Support

Digital nudging refers to the use of user-interface design elements to guide behavior in digital choice environments [6]. In digital menu systems, nudges may appear as personalized suggestions, highlighted combos, default add-ons, popularity labels, or context-aware recommendations. These design elements do not remove customer freedom; instead, they reduce search effort by narrowing the visible choice space and helping customers rely on useful heuristics. Prior work also indicates that well-designed nudges can reduce decision time without necessarily reducing satisfaction [7].

For CrownDine, Apriori-based combos and collaborative-filtering suggestions function as digital nudges. They guide customers toward relevant dishes while keeping the complete menu available. This behavioral interpretation is important because the value of recommendation is not only technical accuracy but also lower cognitive cost during menu exploration.

2.3 Waiting Anxiety and Perceived Control

The customer journey does not end when an order is placed. During the waiting stage, uncertainty about preparation progress can make the wait feel longer and less fair. Waiting psychology suggests that providing timely information can reduce anxiety because customers feel more informed and more in control of the service process. Hui and Zhou [8] show that wait-duration information can increase perceived control, while Dziekan and Kottenhoff [9] find that real-time information can reduce perceived waiting time and improve satisfaction in service contexts.

In restaurant operations, a kitchen display system can therefore serve two purposes. Operationally, it coordinates order preparation. Psychologically, it turns a hidden process into a visible sequence of states, reducing the *black box* effect and helping customers interpret the wait as predictable and fair.

2.4 Association Rule Mining for Food Pairing

Association rule mining is widely used to discover relationships among items in transaction databases. In a restaurant context, each order can be treated as a basket containing one or more dishes. The Apriori algorithm identifies frequent itemsets by repeatedly expanding candidate sets whose support exceeds a minimum threshold. From these itemsets, the system can generate rules such as $X \Rightarrow Y$, meaning that customers who order itemset X also tend to order itemset Y.

Association rules are particularly suitable for combo generation because they are interpretable and directly connected to purchasing behavior. Support indicates how common a combination is, confidence measures the probability of buying the consequent given the antecedent, and lift shows whether the relationship is stronger than random co-occurrence. These measures allow restaurant managers to understand why a combo is recommended and to avoid arbitrary bundling.

2.5 Collaborative Filtering for Personalized Dish Suggestions

Collaborative filtering recommends items by learning from interactions between users and items. User-based collaborative filtering identifies customers with similar ordering histories and recommends dishes liked by similar users. Item-based collaborative filtering identifies dishes that are frequently liked or purchased by similar customer groups. Compared with association rules, collaborative filtering is more personalized because it can recommend items even when they are not part of the same transaction basket.

For menu systems, collaborative filtering can use explicit feedback such as ratings and likes, or implicit feedback such as views, cart additions, purchase frequency, and repeat orders. Because many restaurant systems have sparse rating data, implicit feedback is often more practical. A hybrid design can use Apriori to discover reliable item combinations and collaborative filtering to adapt those combinations to each customer's historical preferences.

2.6 Automatic Combo Design and Business Constraints

Food combos must satisfy both customer preference and restaurant business constraints. A high-quality combo should include complementary dishes, maintain a reasonable price, avoid redundant categories, and remain feasible under inventory conditions. For example, a lunch combo may include one main dish, one side dish, and one drink, while a family combo may include multiple main dishes and shared appetizers.

Automatic combo generation therefore requires more than mining frequent itemsets. The system must filter candidate combos by category rules, discount policy, preparation feasibility, profit margin, and stock availability. Combining data mining with business constraints allows the restaurant to create practical combos that are both behavior-driven and operationally realistic.

3 Methodology

The proposed system transforms user behavior and transaction records into personalized menu recommendations, automatically generated combos, and operational waiting support. The methodology consists of data preprocessing, Apriori rule mining, collaborative filtering, hybrid ranking, combo construction, KDS status synchronization, and evaluation. Figure-level architecture can be interpreted as a dual-phase cognitive support framework: the first phase supports dish choice, and the second phase supports informed waiting.

3.1 Data Collection and Preprocessing

Let $U = \{u_1, u_2, \ldots, u_m\}$ be the set of users and $I = \{i_1, i_2, \ldots, i_n\}$ be the set of menu items. The system collects behavioral events such as menu views, searches, cart additions, purchases, ratings, and repeat orders. Each order transaction is represented as a basket:

$$T_k = \{i_1, i_2, \ldots, i_l\}, \quad T_k \subseteq I. \tag{1}$$

Before model training, raw data are cleaned by removing cancelled orders, merging duplicate item identifiers, normalizing item categories, and filtering inactive menu items. User–item interactions are converted into an implicit preference score:

$$r_{u,i} = w_v V_{u,i} + w_c C_{u,i} + w_p P_{u,i} + w_l L_{u,i}, \tag{2}$$

where $V_{u,i}$ is the number of views, $C_{u,i}$ is the number of cart additions, $P_{u,i}$ is the number of purchases, $L_{u,i}$ is the rating or like signal, and w_v, w_c, w_p, w_l are behavior weights.

3.2 Apriori Association Rule Mining

The Apriori module discovers frequent itemsets from historical order baskets. For an association rule $X \Rightarrow Y$, where $X, Y \subseteq I$ and $X \cap Y = \varnothing$, support, confidence, and lift are defined as:

$$\text{Support}(X \Rightarrow Y) = \frac{|\{T_k \mid X \cup Y \subseteq T_k\}|}{|D|}, \tag{3}$$

$$\text{Confidence}(X \Rightarrow Y) = \frac{|\{T_k \mid X \cup Y \subseteq T_k\}|}{|\{T_k \mid X \subseteq T_k\}|}, \tag{4}$$

$$\text{Lift}(X \Rightarrow Y) = \frac{\text{Confidence}(X \Rightarrow Y)}{\text{Support}(Y)}. \tag{5}$$

Rules are retained when they satisfy minimum support, confidence, and lift thresholds. For example, if customers who order grilled beef frequently order salad and fruit juice, the system can generate a rule that recommends those add-ons when grilled beef is selected. The Apriori output is also used as the candidate pool for automatic combo construction.

3.3 Collaborative Filtering Model

The collaborative filtering module builds a user–item interaction matrix $R \in \mathbb{R}^{m \times n}$, where $r_{u,i}$ represents the preference score of user u for item i. In item-based collaborative filtering, the similarity between two dishes i and j is computed using cosine similarity:

$$\text{sim}(i, j) = \frac{\sum_{u \in U} r_{u,i} r_{u,j}}{\sqrt{\sum_{u \in U} r_{u,i}^2} \sqrt{\sum_{u \in U} r_{u,j}^2}}. \tag{6}$$

The predicted preference of user u for an unseen dish i is calculated as:

$$\hat{r}_{u,i} = \frac{\sum_{j \in N(i)} \text{sim}(i, j) r_{u,j}}{\sum_{j \in N(i)} |\text{sim}(i, j)|}, \tag{7}$$

where $N(i)$ is the set of items similar to i that user u has interacted with. Items with high predicted scores are recommended as personalized suggestions.

3.4 Hybrid Ranking and Recommendation Output

The final recommendation score combines collaborative filtering predictions with association-rule evidence and popularity signals:

$$\text{Score}(u, i) = \alpha \hat{r}_{u,i} + \beta AR(i \mid B_u) + \gamma Pop(i) + \delta Ctx(u, i), \tag{8}$$

where $AR(i \mid B_u)$ is the association-rule score for item i given the user's current basket B_u, $Pop(i)$ is normalized item popularity, $Ctx(u, i)$ represents contextual compatibility such as meal time or category preference, and $\alpha, \beta, \gamma, \delta$ are tunable weights.

The system produces two outputs. The first output is a ranked list of recommended dishes for each user. The second output is a set of suggested add-ons based on the user's current basket, such as drinks, side dishes, or desserts that frequently appear with selected items.

3.5 Automatic Combo Generation

Combo generation starts from frequent itemsets and high-lift association rules. A candidate combo C is defined as:

$$C = \{i_1, i_2, \ldots, i_q\}, \quad q \geq 2. \tag{9}$$

Each candidate combo is validated using category and business constraints:

$$\text{CategoryBalance}(C) = 1, \tag{10}$$

$$P_{min} \leq \text{Price}(C) \leq P_{max}, \tag{11}$$

$$\text{Margin}(C) \geq M_{min}, \tag{12}$$

$$\text{Stock}(i) > 0 \quad \forall i \in C. \tag{13}$$

The combo score is computed as:

$$\text{ComboScore}(C) = \lambda_1 \text{Support}(C) + \lambda_2 \text{Lift}(C) + \lambda_3 \text{Margin}(C) + \lambda_4 \text{Preference}(u, C). \tag{14}$$

Combos with high scores are presented to customers as dynamic bundles, while restaurant administrators can review suggested combos before publishing them as official promotions.

3.6 KDS-Based Waiting Support

After checkout, CrownDine synchronizes order status from the kitchen display system to the customer interface. The order state can be represented as:

$$S_o \in \{\text{Received}, \text{Preparing}, \text{Cooking}, \text{Ready}, \text{Served}\}. \tag{15}$$

For each order o, the interface may display the current state S_o, an estimated remaining time $\hat{t}_o$, and a short progress message. The behavioral objective is to reduce waiting anxiety by changing the customer's perception from uncertain waiting to informed waiting. This component does not necessarily shorten physical preparation time, but it can improve perceived fairness, perceived control, and satisfaction because the customer understands what is happening after ordering.

3.7 Evaluation Metrics

The recommendation quality is evaluated using ranking and classification metrics. Precision@K measures how many of the top-K recommendations are relevant, while Recall@K measures how many relevant items are retrieved:

$$\text{Precision@K} = \frac{|\text{Recommended@K} \cap \text{Relevant}|}{K}, \tag{16}$$

$$\text{Recall@K} = \frac{|\text{Recommended@K} \cap \text{Relevant}|}{|\text{Relevant}|}. \tag{17}$$

Combo quality is evaluated using support, confidence, lift, average order value improvement, click-through rate, and conversion rate. The behavioral and operational layer is evaluated using perceived usefulness, decision time, order-status view rate, perceived waiting time, waiting satisfaction, and customer complaints related to delays. These metrics help determine whether the system improves both customer experience and restaurant revenue.

4 Discussion

The proposed system is suitable for restaurants that want to move from static menu presentation to data-driven personalization and behavior-aware service design. Apriori association rules provide transparent evidence for add-on suggestions and combo creation, making them easy for restaurant managers to interpret. Collaborative filtering complements this by adapting recommendations to each user's previous behavior, which is important when different customers have different tastes, budgets, and ordering habits. From a behavioral perspective, these recommendations operate as digital nudges that reduce decision overload by shrinking the effective choice space without forcing a specific selection.

A hybrid approach is especially useful in restaurant environments because recommendation goals are both customer-oriented and business-oriented. Customers benefit from faster dish discovery and more relevant suggestions, while restaurants benefit from increased cross-selling, improved combo design, and better use of transaction data. In addition, KDS-based status visibility supports the post-order stage by giving customers cognitive control over an otherwise unclear process. The system can also support seasonal promotions by adjusting rule thresholds, context weights, inventory constraints, and operational messages.

Several challenges must be addressed. New users and new menu items may suffer from cold-start problems because there is not enough interaction history. Sparse data can reduce collaborative filtering accuracy, especially for restaurants with limited digital orders. In addition, automatically generated combos must be reviewed carefully to avoid poor taste combinations, excessive discounts, or operational conflicts in food preparation. The waiting-support layer also requires careful design: inaccurate time estimates or excessive status notifications may reduce trust instead of increasing control. Future work should incorporate content-based dish features, nutritional information, real-time A/B testing, and behavioral measurements of decision overload and waiting anxiety.

5 Conclusion

This paper presented CrownDine, a menu recommendation and operational management system that applies Apriori association rule mining and collaborative filtering to suggest dishes, automatically generate food combos, and support customers during the waiting stage. Apriori discovers interpretable co-purchase patterns from transaction baskets, while collaborative filtering personalizes recommendations using user–item interaction data. The hybrid ranking mechanism combines these signals with popularity, context, and business constraints. KDS-based order visibility extends the system beyond menu selection by reducing uncertainty after ordering.

The proposed system provides a practical foundation for intelligent digital menus. By recommending relevant dishes and constructing behavior-driven combos, restaurants can reduce customer decision effort, increase average order value, and improve the effectiveness of promotions. By showing order progress, the system can also reduce waiting anxiety and improve perceived fairness even when actual preparation time remains unchanged. Overall, CrownDine demonstrates that restaurant technology should be evaluated not only by algorithmic performance but also by its ability to reduce cognitive friction across the customer journey. Future development will focus on cold-start handling, richer user profiling, nutritional recommendation constraints, waiting-time prediction, and deployment evaluation in real restaurant ordering systems.

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